

EXERCISE 1 (2.5 POINTS).

Execution:

a (0.5)	b (0.5)	c (0.5)	j (0.5)	Ecran (0.5)
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0	3	15		
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				c = 15
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1		14	1	
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3		13	2	
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			3	$b = 3$
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	6		4	$j = 4$
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				$c = 13$
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4		12	1	
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6		11	2	
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			3	$b = 6$
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	5		4	$i = 4$
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				3 - 6

EXERCISE 2 (3 POINTS).

Write an algorithm (in algorithmic language) that calculates and displays the n th term (U_n , $n > 3$) of the Fibonacci sequence defined by: $U_1=0$; $U_2= 1$; $U_3=2$ et $U_{n+1}=U_n - 3 U_{n-2} + 5$.

- The value of N is read from the keyboard

<i>Comment</i>	ALGORITHM EXO2 (solution 1)	Solution 2
<i>Declaration</i> (0.5)	n, U1, U2, U3, U, i: Integer	
	BEGIN	
<i>Reading</i> (0.25)	Do scan(n); while (n<=3);	
<i>Initialization</i> (0.5)	$U1 \leftarrow 0$; $U2 \leftarrow 1$; $U3 \leftarrow 2$; $U \leftarrow U3$;	
<i>Treatment</i> (1.5) <i>tel que:</i> <i>Boucle (0.25)</i> <i>Formule de récurrences (0.25)</i> <i>Permutation entre U_i (1 point)</i>	for $i \leftarrow 4$ to n do begin $U \leftarrow U3$; //U temporaire variable $U3 \leftarrow U3 - 3 * U1 + 5$; $U1 \leftarrow U2$; $U2 \leftarrow U$; END	for $i \leftarrow 4$ to n do BEGIN //we can use switch if (i mod 3 = 1) then Begin $U \leftarrow U - 3 * U1 + 5$; $U1 \leftarrow U$; End if (i mod 3 = 2) then Begin $U \leftarrow U - 3 * U2 + 5$; $U2 \leftarrow U$; End if (i mod 3 = 0) then Begin $U \leftarrow U - 3 * U3 + 5$; $U3 \leftarrow U$; End END
	for $i \leftarrow 4$ to n do begin $U \leftarrow U3 - 3 * U1 + 5$; $U1 \leftarrow U2$; $U2 \leftarrow U3$; $U3 \leftarrow U$; END	
<i>Writing</i> (0.25)	<code>printf("U ", n , "=" , U3);</code>	<code>printf("U ", n , "=" , U);</code>
	END.	